

Minimal block V: the freezing inequality

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Abstract

For the block state integral with general jointly continuous ξ, η (block $u \in (0, b]^m$, non-critical $v \in (0, b]^{d'}$) and its frozen version with $\xi(0, v), \eta(0, v)$, the mean-value bound and the corner/strip split of unit 76 give, at every scale $N \geq 0$, $|Z(N) - Z_0(N)| \leq C_2(\omega Z_h^{\beta/2, 2L}(N) + \frac{2M+2L}{\delta} \sum_i Z_{h+e_i}^{\beta/2, 2L}(N))$, where ω bounds the oscillation of ξ, η on the corner $\{u_i \leq \delta\}$ and the right-hand side consists of constant- ξ state integrals. Zero sorry/axiom.

1 The things formalised

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noncomputable def blockStateIntegral ( b N : ) {m d' : } (k h : Fin m → )
  (k' h' : Fin d' → ) ( : (Fin m → ) → (Fin d' → ) → ) : :=
  v, ( i, v i ^ h' i) * ( u, ( i, u i ^ h i) * u v
    * quadKernel ( u v) (N * ( i, v i ^ k' i) * i, u i ^ k i) (boxMeasure b m))
    (boxMeasure b d')
noncomputable def constStateIntegral ( a b N : ) ... : :=
  v, ( i, v i ^ h' i) * boxIntegralFin a b (N * i, v i ^ k' i) k h
    (boxMeasure b d')
theorem constStateIntegral_eq_frozen ... :
  constStateIntegral a b N k h k' h'
    = frozenStateIntegral b N (toNatFun k 1) (toNatFun h 0) d k' h'
      (fun _ => a) (fun _ => 1)
theorem blockStateIntegral_sub_le ( b : ) (h : 0 < ) (hb : 0 < b) ... (L M : ) ...
  ( : ) (h : 0 ) (h : 0 < )
  (hcorner : u v, ... → ( i, u i ) → | u v - 0 v| + | u v - 0 v| )
  (N : ) (hN : 0 N) :
  |blockStateIntegral b N k h k' h'
    - blockStateIntegral b N k h k' h' (fun _ v => 0 v) (fun _ v => 0 v)|
    max 1 (M * Real.exp ( / 2))
    * ( * constStateIntegral ( / 2) (2 * L) b N k h k' h'
      + (2 * M + 2 * L) /
        * i, constStateIntegral ( / 2) (2 * L) b N k (shiftExp h i) k' h')

```

2 Proof strategy

Everything is done on the product box measure, where all integrands are continuous on the compact closed box, hence integrable (`integrable_box_prod_of_continuous`); `integral_prod` converts to the iterated forms. On the open product box (almost everywhere) the integrand difference is $v^{h'} u^h |\eta f_\xi - \eta_0 f_{\xi_0}|(s)$, bounded by $v^{h'} u^h C_2 D e^{-(\beta/2)s^2 + \beta L s}$ and then, via the corner/strip split, by $C_2(\omega + \frac{D_{\max}}{\delta} \sum_i u_i) v^{h'} u^h e^{-(\beta/2)s^2 + \beta L s}$; the terms $u^h u_i = u^{h+e_i}$ are recognised as shifted-exponent

integrands. Then $\|\int(G - G_0)\| \leq \int\|G - G_0\| \leq \int H$ and $\int H$ is expanded by linearity into the constant- ξ state integrals.

3 Roadblocks and resolutions

- `ae_restrict_mem` on a product of pi measures needs both `volumes` and the set given explicitly; otherwise the measurable-space instance is left as a metavariable.
- `integral_congr_ae` pointwise goals are beta redexes; `beta_reduce` before `rw`.
- The unifier cannot recover `fun _ v => 0 v` from a continuity proof; pass the frozen functions explicitly to `blockIntegrand_continuous`.
- The bound $|\xi(0, v)| \leq L$ needs $(0, v)$ in the closed box, hence $0 \leq b$: add `hb : 0 < b`.

4 Outcome

The freezing inequality reduces the general multiplicity- m leading term to the constant- ξ theory already formalised. Next: the limit argument ($N^p \log^{-d} N$ -strip terms $\rightarrow 0$, then an ε -argument with uniform continuity on the closed box) and the main theorem $Z \sim C N^{-p} \log^d N$.